

MEDICAL INSURANCE COST

Statistical Analysis & Interactive Dashboard

Lab-4: Applied Statistical Modeling & Interactive Web Dashboard

Course: Statistical Modeling with Python

Course Code: DS602

Program: M.Sc. Data Science

Semester: 1

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1. Project Overview

This project applies statistical methods to the Medical Insurance Costs dataset and delivers the analysis through an interactive Streamlit dashboard. The work combines exploratory data analysis, descriptive statistics, hypothesis testing, multiple linear regression, and regression diagnostics. The dashboard makes statistical results accessible through interactive filters, visualizations, test controls, and live prediction.

The project follows the DS602 Statistical Methods Lab-4 requirements for Applied Statistical Modeling & Interactive Web Dashboard.

Dashboard: [Interactive Streamlit Dashboard](#)

2. Project Objectives

- Explore the structure and distribution of medical insurance cost data.
- Calculate mean, median, standard deviation, IQR, skewness, and kurtosis.
- Use distribution plots, bivariate plots, and correlation analysis to study relationships.
- Perform a two-group hypothesis test with Shapiro-Wilk and Levene assumption checks.
- Use a two-sample t-test when appropriate or Mann-Whitney U when normality is not appropriate.
- Perform a Chi-Square test of independence between categorical variables.
- Fit a multiple linear regression model using statsmodels OLS.
- Interpret coefficients, p-values, 95% confidence intervals, R^2 , and adjusted R^2 .
- Check residual assumptions using residual-vs-fitted plots, Q-Q plots, Jarque-Bera/Omnibus testing, and VIF.
- Provide interactive data exploration, hypothesis testing, prediction, and diagnostics in Streamlit.

3. Dataset

The project uses the Medical Insurance Costs dataset, containing information about individuals and their medical insurance charges. The dataset has 1,338 observations and 7 variables

Variable	Type	Description
age	Numerical	Age of primary beneficiary
sex	Categorical	Sex of primary beneficiary
bmi	Numerical	Body Mass Index
children	Count	Number of children/dependents
smoker	Categorical	Smoking status
region	Categorical	Geographic region

charges	Numerical	Medical insurance charges
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4. Technologies and Libraries

Python, Pandas, NumPy, Matplotlib, Seaborn, SciPy, Statsmodels, Streamlit, Git, GitHub, Streamlit Community Cloud, Visual Studio Code

Library/Tool	Purpose	Usage
Python	Programming	Core implementation
Pandas	Data handling	Loading, cleaning, filtering, summaries
NumPy	Numerical computing	Numerical operations
SciPy	Statistical testing	Shapiro-Wilk, Levene, t-test/Mann-Whitney, Chi-Square
statsmodels	Statistical modeling	OLS, confidence intervals, residual tests, VIF
Streamlit	Interactive web app	Dashboard interface
Plotly	Interactive plots	Reactive visualizations
Seaborn	Statistical plots	EDA visualizations
Matplotlib	Plotting	Diagnostic plots

5. Project Structure and Environment

```
Medical-Insurance-Dashboard/  
├── data/  
│   └── insurance.csv  
├── analysis.py  
├── app.py  
├── requirements.txt  
├── README.md  
└── .gitignore
```

- 1. Create the virtual environment: `python -m venv venv`
- 2. Activate the virtual environment.
- 3. Install packages: `pip install -r requirements.txt`
- 4. Run the application: `streamlit run app.py`
- 5. Open the Streamlit application at `localhost:8501`.

6. Exploratory Data Analysis

6.1 Data Preparation

The CSV is loaded with Pandas. The workflow examines dimensions, data types, missing values, duplicate observations, and summary information before statistical analysis.

- Inspect rows and columns.
- Check data types and missing values.
- Check duplicate observations.
- Separate numerical and categorical variables.

6.2 Descriptive Statistics

For numerical variables, the project calculates the central tendency and dispersion measures required by the assignment.

- Mean
- Median
- Standard deviation
- Interquartile range (IQR)
- Skewness
- Kurtosis

6.3 Visual Exploration

- Histograms/KDE plots for numerical distributions.
- Bivariate scatter plots for numerical relationships.
- Correlation matrix/heatmap.
- Categorical comparisons of medical charges across groups.

7. Hypothesis Testing

7.1 Two-Group Comparison

A continuous metric such as medical charges is compared between two groups, with smoker versus non-smoker being a natural example for this dataset.

H0: There is no significant difference between the two groups.

H1: There is a significant difference between the two groups.

Significance level: $\alpha = 0.05$.

6. Use Shapiro-Wilk to assess normality.
7. Use Levene's test to assess equality of variances.
8. If normality is appropriate, use a two-sample t-test.
9. If normality is not appropriate, use the Mann-Whitney U test.
10. Compare the resulting p-value with 0.05 and state Reject H0 or Fail to Reject H0.

7.2 Chi-Square Test of Independence

The second hypothesis test examines whether two categorical variables are statistically associated, for example smoking status and region.

H0: The categorical variables are independent.

H1: The categorical variables are associated.

Decision rule: $p < 0.05 \rightarrow$ reject H0; $p \geq 0.05 \rightarrow$ fail to reject H0.

8. Multiple Linear Regression

Medical charges are modeled as the response variable using multiple explanatory variables. Categorical predictors are encoded appropriately before fitting the OLS model.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \epsilon$$

- Coefficients estimate the effect of predictors while holding other predictors constant.
- p-values assess evidence that individual coefficients differ from zero.
- 95% confidence intervals provide uncertainty ranges for coefficient estimates.
- R^2 measures the proportion of response variation explained by the model.
- Adjusted R^2 accounts for model complexity.

9. Regression Diagnostics

- Residuals versus fitted values: assesses linearity and homoscedasticity.
- Q-Q plot: assesses approximate normality of residuals.
- Jarque-Bera or Omnibus test: provides a formal residual-normality check.
- Variance Inflation Factor (VIF): checks multicollinearity among continuous predictors.

10. Interactive Streamlit Dashboard

Dashboard Area	Features	Purpose
Tab 1 – Data Exploration	Sidebar filters, category selections, summaries, distributions, scatter plots, correlation	Interactive EDA
Tab 2 – Hypothesis Testing Lab	Dynamic variable selection, test statistic, p-value, conclusion	Interactive inference
Tab 3 – Live Prediction & Diagnostics	User inputs, prediction, 95% interval, residual diagnostics	Prediction and model validation

11. Statistical Findings and Interpretation

The numerical findings reported in the final submission should be taken directly from the outputs generated by analysis.py and the Streamlit dashboard. No numerical values are invented in this documentation.

- Descriptive statistics summarize the center, variability, skewness, and kurtosis of numerical features.
- Distribution plots identify skewness and unusual/extreme observations.
- Correlation analysis identifies linear relationships between numerical variables.
- The two-group test determines whether the selected continuous metric differs significantly between the selected groups.
- The Chi-Square test evaluates association between selected categorical variables.
- OLS evaluates the joint relationship between medical charges and multiple predictors.
- Regression diagnostics indicate whether model assumptions are reasonably satisfied.
- VIF identifies potential multicollinearity among continuous predictors.
- The prediction interface demonstrates model-based estimation for user-supplied characteristics.

12. Reproducibility

11. Open the project folder in VS Code.
12. Verify that data/insurance.csv is present.
13. Activate the virtual environment.
14. Install dependencies from requirements.txt.
15. Run analysis.py when statistical outputs are required.
16. Launch the dashboard with streamlit run app.py.
17. Use the three dashboard areas to explore, test, model, and predict.

13. Version Control and Submission

The project is maintained in a GitHub repository with the source code, dataset folder, dependencies, README, and configuration files.

- app.py – Streamlit dashboard.
- analysis.py – statistical analysis and modeling.
- requirements.txt – dependencies.
- data/insurance.csv – dataset.
- README.md – project documentation.
- .gitignore – excludes environment/cache files.

The final academic documentation is converted to PDF and added to the same GitHub repository. The repository link can then be submitted through Google Classroom.

14. Limitations

- The data are observational; statistical association does not establish causation.
- Medical charges may depend on factors not represented in the dataset.
- Categorical reference coding affects regression coefficient interpretation.
- OLS inference depends on assumptions that must be assessed using diagnostics.
- Predictions and intervals represent statistical estimates and contain uncertainty.

15. Conclusion

This project demonstrates an end-to-end statistical workflow using the Medical Insurance Costs dataset. It integrates descriptive statistics, exploratory visualization, formal hypothesis testing, multiple linear regression, diagnostic checking, and interactive prediction in a Streamlit application. The structured project also demonstrates reproducible development through requirements management, source-code separation, README documentation, and GitHub version control.

Appendix A – Statistical Decision Rules

Test	Null Hypothesis	Decision at $\alpha=0.05$
Shapiro-Wilk	Data/residuals are normally distributed	$p < 0.05 \rightarrow \text{reject } H_0$
Levene	Group variances are equal	$p < 0.05 \rightarrow \text{reject } H_0$
Two-sample t-test	Group means are equal	$p < 0.05 \rightarrow \text{reject } H_0$
Mann-Whitney U	Group distributions are equivalent	$p < 0.05 \rightarrow \text{reject } H_0$
Chi-Square	Categorical variables are independent	$p < 0.05 \rightarrow \text{reject } H_0$
Regression coefficient	$\beta_j = 0$	$p < 0.05 \rightarrow \text{reject } H_0$